Assignment 5: Final Competition

Dr. Joseph Conroy (conroyj@umd.edu) 80 points
Due: Tuesday, 11 MAY, 2026 @ 11:59 PM

1 The Challenge

This end-of-year competition asks each team to apply what it has learned in ENAE450 on the physical TurtleBot 4. Your task is to deploy a fully autonomous ROS 2 stack that navigates a box-built maze from the **Start Zone** to the **Finish Zone**. The primary sensor will be the onboard 2D LiDAR. You may use any approach covered in class, as well as additional methods you research independently.

There will be **two races**: one with a short pre-mapping opportunity and one in a novel, unmapped environment. Each team may attempt up to **three official runs per race**, and the best time will count. The competition will take place in the 3119 IDEA Factory RAL laboratory.

2 Race Design

1) Race with Pre-Mapping Opportunity

Teams will receive **one 3-minute teleoperation window** immediately before the official run to generate and save a map, for example with `map_saver_cli`. Each team will be able to verify

network connectivity as well prior to the start of the official runs.

The maze will be moderately complex but identical for all teams. The robot will start from the same location for every run, and the finish zone will remain fixed relative to the start.

The official run begins immediately after the mapping phase. Teams must then rely on their saved map for navigation. The TurtleBot will start at the enclosed end of the maze with a random orientation. A run is complete when the robot exits the maze, turns 180 degrees, and comes to a stop.

Wall contact or intervention results in a time penalty. Teams must prepare a single launch file or startup script that begins the run without manual intervention. Once your launch file or startup script is executed, **no human intervention** is allowed during the official run, including teleoperation by controller or keyboard. Primary processing should run across the TurtleBot 4 Raspberry Pi and your Fast DDS-connected host PC on the `RAL_robots` lab network. No additional hardware or software changes are permitted on the TurtleBot 4 platforms.

Conduct

- Teams must remain within the established "operator zone" during official runs.
- Only the assigned team may interact with the robot in the run area.

2) Novel Environment Race

This race follows the same rules as the pre-mapping race, except teams will have **no opportunity** to map the environment in advance.

Deliverables:

- Create a folder named "Final_Competition" in one team member's repository.
- In that folder, create "final_competition_data.md" and list the team members.
- Upload a clear video of your best autonomous maze run and place its Google Drive link under "video:" in "final_competition_data.md".
- Upload the official MCAP ROS 2 bag and place its Google Drive link under "ros2_bag:" in "final_competition_data.md".
- Include the TurtleBot 4 ROS 2 package source code used for the competition in the submission repository.
- Submit a PDF technical report as described below.

final_competition_data.md example:

```
team_members: Rusty Jointz, Anne Droid, Lou P. Back
video: https://drive.google.com/your_video_link_here
ros2_bag: https://drive.google.com/your_ros2_bag_link_here
```

Final report: Use the final report to present your design process, results, and lessons learned. Spelling and grammar matter. Upload the PDF to ELMS and place a copy in your repository alongside "final_competition_data.md". You may add other sections, but the report must include the following:

- Introduction
- Hardware: Methods, Results
- Top-down run plot showing robot position (x versus y) with clear labels, a legend, and a caption
- Retrospective: what worked, what did not, and what you would change
- Team contribution summary describing who did what

Include figures or short code snippets as needed to explain your process, and make sure all figures have captions and labels. Do not include large code listings or a full code appendix; the repository already serves that purpose.

3 Evaluation Metrics and Scoring Rubric

Your grade and competition standing for each race type are determined by three primary metrics.

3.1 Metric A: Time to Completion (10 points premapping; 20 points novel environment)

Primary performance metric: Measured by stopwatch for each race type from when the robot starts moving to when the entire robot completes the final 180-degree maneuver and stops.

A.1 Base Time

Stopwatch measured time as described above with no penalties.

A.2 Penalties

- **Collision penalty:** +CP (TBD) seconds for each distinct wall/box collision.
- **Intervention penalty:** +IP (TBD) seconds per teleoperation or physical intervention when the robot is stuck.

"TBD" here means we will gather initial data from the first few runs and then set these penalty values. These values will be chosen to discourage penalties. That is, "rushing" to the finish by plowing through boxes will **not** be a winning strategy.

A.3 "Did Not Finish" Rule

If the robot cannot complete the maze after **3 interventions** or **5 minutes**, the team will be assigned **DNF** for Metric A. The team can still earn credit in Metrics B and C.

3.2 Metric B: Novelty and Professionalism (Subjective/Technical, 20 points)

- **Code architecture and professionalism (10 pts):** modular ROS 2 package design, clean launch structure, parameters, naming, version control hygiene.
- **Algorithmic novelty (10 pts):** meaningful additions beyond tutorial defaults (sensor fusion, custom planners, custom costmap behavior, adaptive logic, etc.).

3.3 Metric C: Data Logging and Post-Run Analysis (Deliverable, 30 points)

- **MCAP ROS 2 bag (5 pts):** must include at minimum `/tf`, `/odom`, `/scan`, and `/cmd_vel`.
- **Post-run visualization/report (25 pts):** As discussed in the Final Report section, this includes a clear top-down plot of the robot's path.

Tie-Breaking Rule

If two teams have identical adjusted Metric A times, tie-break in this order:

1. Fewer collisions,
2. Fewer interventions,
3. Higher Metric B score.

4 Scoring Summary Table

Category	**Points**
Metric A: Time to Completion (with penalties)	30
Metric B: Novelty and Professionalism	20
Metric C: Data Logging and Post-Run Analysis	30
Total	**80**